external.”3 Gottfried Leibniz and others vociferously disagreed, claiming that space and time are merely bookkeeping devices for conveniently summarizing relationships between objects and events within the universe. The location of an object in space and in time has meaning only in comparison with another. Space and time are the vocabulary of these relations, but nothing more. Although Newton’s view, supported by his experimentally successful three laws of motion, held sway for more than two hundred years, Leibniz’s conception, further developed by the Austrian physicist Ernst Mach, is much closer to our current picture. As we have seen, Einstein’s special and general theories of relativity firmly did away with the concept of an absolute and universal notion of space and time. But we can still ask whether the geometrical model of spacetime that plays such a pivotal role in general relativity and in string theory is solely a convenient shorthand for the spatial and temporal relations between various locations, or whether we should view ourselves as truly being embedded in something when we refer to our immersion within the spacetime fabric.

Although we are heading into speculative territory, string theory does suggest an answer to this question. The graviton, the smallest bundle of gravitational force, is one particular pattern of string vibration. And just as an electromagnetic field such as visible light is composed of an enormous number of photons, a gravitational field is composed of an enormous number of gravitons—that is, an enormous number of strings executing the graviton vibrational pattern. Gravitational fields, in turn, are encoded in the warping of the spacetime fabric, and hence we are led to identify the fabric of spacetime itself with a colossal number of strings all undergoing the same, orderly, graviton pattern of vibration. In the language of the field, such an enormous, organized array of similarly vibrating strings is known as a coherent state of strings. It’s a rather poetic image—the strings of string theory as the threads of the spacetime fabric—but we should note that its rigorous meaning has yet to be worked out completely.

Nevertheless, describing the spacetime fabric in this string-stitched form does lead us to contemplate the following question. An ordinary piece of fabric is the end product of someone having carefully woven